

Assignment 2:

Part 1: INNER JOIN

Q1. Retrieve a list of all orders along with the customer's company name.

```
Query  Query History
1  SELECT
2      ord.order_id AS order_id
3      , cust.company_name AS company_name
4  FROM orders AS ord
5  INNER JOIN customers as cust
6  ON ord.customer_id=cust.customer_id
```

Q2. Get the order details (OrderID, ProductName, UnitPrice, Quantity) by joining the Orders and OrderDetails tables.

```
Query  Query History
1  SELECT
2      ord.order_id AS order_id,
3      ord.unit_price AS unit_price,
4      ord.quantity AS quantity,
5      prod.product_name AS product_name
6  FROM order_details AS ord
7  INNER JOIN products AS prod
8  ON ord.product_id=prod.product_id
```

Q3. Fetch all employees and their respective territories from the Employees and EmployeeTerritories tables.

```
Query  Query History
1  SELECT
2      emp.first_name AS first_name,
3      emp.last_name AS last_name,
4      trt.territory_description AS territory_desc
5  FROM employees AS emp
6  INNER JOIN employee_territories AS emp_t
7  ON emp.employee_id=emp_t.employee_id
8  INNER JOIN territories trt
9  ON emp_t.territory_id=trt.territory_id
```

Part 2: LEFT JOIN

Q1. Retrieve a list of all customers and their orders, including those who haven't placed any orders.

```
Query  Query History
1  SELECT cust.contact_name as contact_name
2  FROM customers AS cust
3  LEFT JOIN orders AS ord
4  ON cust.customer_id=ord.customer_id
```

Q2. Display a list of products along with the total quantity sold, including products that have never been ordered.

Query	Query History
1	SELECT
2	prod.product_id AS product_id,
3	prod.product_name AS product_name,
4	SUM(ord.quantity) AS total_quantity
5	FROM products AS prod
6	LEFT JOIN order_details AS ord
7	ON prod.product_id=ord.product_id
8	GROUP BY prod.product_id, prod.product_name
9	

Q3. Get all suppliers and their products, including suppliers that don't currently supply any products.

Query	Query History
1	SELECT
2	prod.product_name AS product_name,
3	supl.company_name AS company_name
4	FROM products AS prod
5	LEFT JOIN suppliers AS supl
6	ON prod.supplier_id=supl.supplier_id
7	

Part 3: RIGHT JOIN

Q1. Retrieve a list of all orders along with the customer's contact name, ensuring all orders are included, even if the customer's contact name is missing.

Query	Query History
1	SELECT
2	ord.order_id,
3	cust.contact_name
4	FROM orders AS ord
5	RIGHT JOIN customers AS cust
6	ON ord.customer_id=cust.customer_id

Q2. Fetch all orders and their shippers, ensuring all orders are included, even if no shipper is assigned.

Query	Query History
1	SELECT
2	ord.order_id AS order_id,
3	ship.company_name AS shipper_name
4	FROM orders AS ord
5	RIGHT JOIN shippers AS ship
6	ON ord.ship_via=ship.shipper_id

Part 4: FULL OUTER JOIN

Q1. Retrieve a list of all customers and suppliers, including those who have no corresponding records in the other table.

Query	Query History
1	SELECT
2	cust.contact_name AS customer_contact_name,
3	supl.company_name AS supplier_company_name
4	FROM customers as cust
5	FULL OUTER JOIN orders AS ord
6	ON cust.customer_id = ord.customer_id
7	FULL OUTER JOIN order_details AS ord_d
8	ON ord.order_id=ord_d.order_id
9	FULL OUTER JOIN products as prod
10	ON ord_d.product_id=prod.product_id
11	FULL OUTER JOIN suppliers as supl
12	ON prod.supplier_id=supl.supplier_id
13	

Q2. Get a list of all employees and their orders, including employees who haven't processed any orders and orders that don't have an assigned employee.

Query	Query History
1	SELECT
2	emp.first_name,
3	emp.last_name,
4	ord.order_id
5	FROM employees AS emp
6	FULL OUTER JOIN orders AS ord
7	ON emp.employee_id=ord.employee_id
8	

Q3. Retrieve all products and their order details, ensuring that products with no orders and orders missing product details are also included.

Query	Query History
1	SELECT
2	prod.product_name,
3	ord_d.order_id
4	FROM products AS prod
5	FULL OUTER JOIN order_details AS ord_d
6	ON prod.product_id=ord_d.product_id
7	
8	

SELF JOIN & CROSS JOIN

Q1. Find all pairs of employees who work in the same city (use a SELF JOIN).

Query	Query History
1	SELECT
2	emp1.first_name AS employee1,
3	emp2.first_name AS employee2,
4	emp1.city AS city
5	FROM employees AS emp1
6	JOIN employees AS emp2
7	ON emp1.city=emp2.city

Q2. Retrieve a cross join of all categories and suppliers, showing every possible combination.

Query	Query History
1	SELECT
2	catg.category_name AS category_name,
3	supl.company_name AS company_name
4	FROM categories AS catg
5	CROSS JOIN suppliers AS supl;